

CS101

Scope & Do-While Loops

3/10

Announcements

1. Lab6 (worksheet) due Friday 3/13 at midnight
2. Extended HW3 (Mario) to Friday 3/20 at midnight
 - a. Start soon!
3. Added to vocab:
 - a. Debugger
 - b. Hard-coded
 - c. Scope

Agenda

1. Lab8 Do-While
2. Scope Demo (with debugger!)
 - a. No Lab for scope
3. Quiz7 Kahoot
4. Start Lab9 Guess Game
5. Open work time

SCOPE IN JAVA

SCOPE The region of code where a variable exists and can be accessed. A variable declared inside a block `{ }` only exists within that block.

• IF/ELSE – OUT OF SCOPE

```
int x = 10;
if (x > 5) {
    int y = 20;
}
System.out.println(x); // OK!
System.out.println(y); // ERROR
```

y was declared inside the if block – it doesn't exist outside of it.

• IF/ELSE – FIXED

```
int x = 10;
int y = 0; // declare BEFORE
if (x > 5) {
    y = 20; // assign inside
}
System.out.println(x); // OK!
System.out.println(y); // OK!
```

Declared before the block, so it still exists after.

• FOR LOOP – OUT OF SCOPE

```
for (int i = 0; i < 5; i++) {
    System.out.println(i);
}
System.out.println(i); // ERROR
```

i is declared in the loop header – it only lives inside the loop.

• FOR LOOP – FIXED

```
int i; // declare BEFORE
for (i = 0; i < 5; i++) {
    System.out.println(i);
}
System.out.println(i); // OK! (prints 5)
```

Declared before the loop so it survives after the loop ends.

Demo scope.java